

MATHEMATICS

MATHEMATICS GRADE 12 TERM 3 CONTROLLED ASSESSMENT PACK

SUBJECT	MATHEMATICS
GRADE	12
ASSESSMENT TASK	MATHEMATICS GRADE 12 TERM 3 CONTROLLED ASSESSMENT PACK
MARKS	50
TIME	1 hour

Educator approval required before classroom use.

INSTRUCTIONS AND INFORMATION

1. Read all instructions carefully.
2. Answer all questions or complete all activities as instructed.
3. Show all working where calculations, planning, diagrams or explanations are required.
4. Use the space provided by the educator, answer booklet or learner task book.
5. Write neatly and clearly.
6. Do not use a calculator unless the educator allows it.

SOURCE MATERIAL

SOURCE 1: Official-paper exemplar: Diagram Or Model

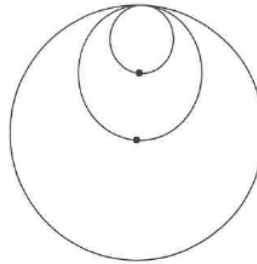
Mathematics/P1

4
NSC Confidential

DBE/November 2024

QUESTION 3

A circle with radius 6 cm is drawn.
A second, smaller circle is drawn through the centre of the first circle and also touches the first circle internally, as shown in the diagram.
A third, smaller circle is drawn through the centre of the second circle and touches the second circle internally. The process of drawing circles continues and forms a geometric pattern.



- 3.1 Write down the radius of the 3rd circle. (2)
- 3.2 Calculate the sum of the areas of the first 10 circles. (4)
- 3.3 Which circle has a diameter of $\frac{3}{128}$ cm? (4)
- [10]

QUESTION 4

Given: $f(x) = a^x - 1$ for $a > 0$. $B\left(2; -\frac{5}{9}\right)$ is a point on f .

- 4.1 Calculate the value of a . (2)
- 4.2 Write down the range of f . (1)
- 4.3 Sketch the graph of f . Clearly show the intercepts with the axes and asymptotes, if any. (3)
- 4.4 It is further given that C is a point on f at $y = \frac{19}{8}$.

Determine the coordinates of C' , the image of C, when C is reflected about the line $y = x$. (3)

[9]

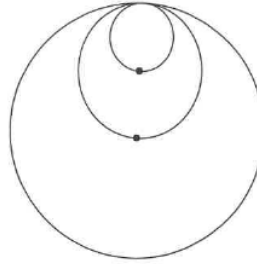
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SOURCE 2: Official-paper exemplar: Graph Or Chart

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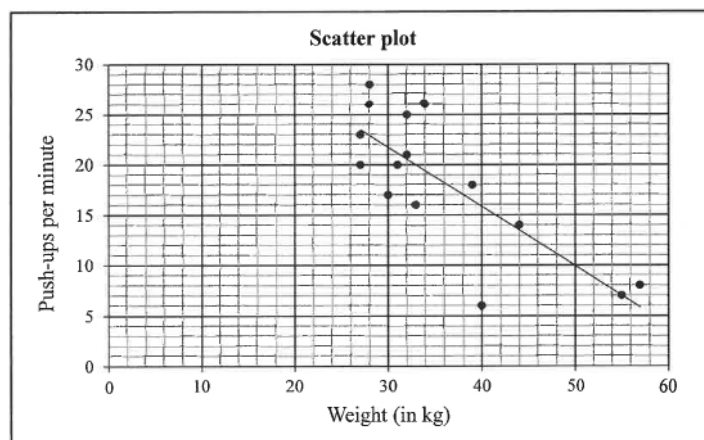
[9]

SOURCE 3: Official-paper exemplar: Data Table

QUESTION 1

At the beginning of a season, the coach of a junior boys' rugby team recorded the weight (in kg) of the 15 players in his team and the number of push-ups that each player was able to do in one minute. The data is represented in the table and scatter plot below. The least squares regression line for the data is drawn. ~

Weight (in kg) (x)	34	32	40	27	33	28	27	55	39	44	30	57	28	32	31
Number of push-ups per minute (y)	26	21	6	20	16	26	23	7	18	14	17	8	28	25	20



- 1.1 Determine the equation of the least squares regression line for the data. (3)
- 1.2 Write down the correlation coefficient. (1)
- 1.3 The coach uses the least squares regression line to set the target for the minimum number of push-ups by each team member according to their weight. Predict the number of push-ups that a member of the team, who weighs 29 kg, should do to meet the target. (2)
- 1.4 Write down the mean number of push-ups for the given data. (1)
- 1.5 The players trained hard during the season. At the end of the season, the coach reported that each player was able to do 5 more push-ups per minute than they did at the beginning of the season. How does the increase in the number of push-ups influence the standard deviation of the data? (1)
- 1.6 At the beginning of the season, the coach used the least squares regression line as the minimum target for a player to aim for. Determine the maximum possible increase in the number of push-ups that a team member must obtain to reach the minimum target. (2)

[10]

QUESTIONS

QUESTION 1: SECTION A: CONCEPTS AND PROCEDURES

Instructions: Show all working.

Stimulus: Topic focus: statistics, probability, Euclidean geometry and measurement.

1.1. Solve a routine problem linked to the stated term focus and show all algebraic working. **(6)**

1.2. Interpret the supplied graph, table or diagram and state the meaning of one key feature. **(6)**

1.3. Complete a multi-step calculation using the supplied values and units. **(8)**

QUESTION 2: SECTION B: PROBLEM SOLVING

Instructions: Use the workspace and justify each step.

Stimulus: Problem context: data displays, probability rules and geometric proof.

2.1. Set up and solve a non-routine problem from the term context. **(10)**

2.2. Justify one method step or proof statement. **(5)**

QUESTION 3: SECTION C: INTEGRATED APPLICATION

Instructions: Combine concepts from the term pack.

Stimulus: Integrated focus: mixed-topic source-based calculation paper.

3.1. Answer an integrated application question and comment on the reasonableness of the result. **(15)**
